WELD MONITOR SYSTEM

Project Report

STM32F446RE Nucleo Board | RIGOL DG4162 Function Generator

Date: March 30, 2026

1. Project Overview

The Weld Monitor System is an embedded application developed on the STM32F446RE Nucleo board using STM32CubeIDE and STM32CubeMX. The system reads an analogue weld signal via ADC, performs real-time signal processing, and detects four types of weld defects:

- Spatter (SP) — High standard deviation of voltage signal
- Burn-through (BU) — High RMS voltage exceeding threshold
- Instability (IN) — High voltage range (Max - Min)
- Porosity (PO) — Voltage minimum below threshold (low signal)

A RIGOL DG4162 Function/Arbitrary Waveform Generator was used to simulate weld signals for testing. Results were monitored in real-time via UART (PuTTY terminal at 115200 baud).

2. Hardware Setup

2.1 Components Used

Component	Details
Microcontroller	STM32F446RETx (NUCLEO-F446RE)
IDE	STM32CubeIDE 2.1.1
Config Tool	STM32CubeMX
Function Generator	RIGOL DG4162 (160 MHz, 500 MSa/s)
ADC Pin	PA0 (ADC1 Channel 0)
UART	USART2 @ 115200 baud
LED Indicator	LD2 (PA5) — ON during fault
Terminal	PuTTY on COM10

2.2 Hardware Connection

Function Generator CH1 output was connected to the Nucleo board as follows:

- CH1 Signal (+) → A0 pin (PA0) on Nucleo board
- CH1 GND → GND pin on Nucleo board
- Output impedance set to HighZ to avoid voltage division

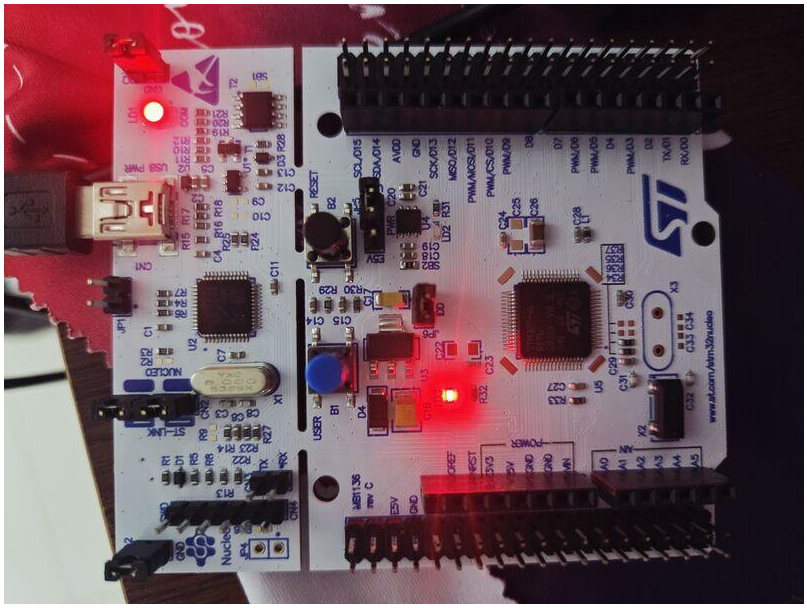

Fig 1: STM32F446RE Nucleo Board Setup

3. Software Architecture

3.1 Detection Thresholds

Fault Type	Parameter	Threshold	Condition
Spatter (SP)	STD Dev	2.1943 V	STD > threshold
Burn-through (BU)	RMS	2.1607 V	RMS > threshold
Instability (IN)	Range (Max-Min)	0.2410 V	Range > threshold
Porosity (PO)	MIN voltage	0.8000 V	MIN < threshold

3.2 Signal Processing Pipeline

- ADC samples at 100 Hz (10ms per sample) via HAL_ADC_PollForConversion
- Circular buffer of 100 samples (WINDOW_SIZE = 100)
- RMS, STD, MIN, MAX computed on full window
- Fault flags set based on threshold comparison
- Results transmitted via USART2 in formatted string
- LD2 LED turns ON if any fault is detected

4. Test Results

4.1 Test 1 — High Amplitude Signal (Burn-through + Instability)

Function Generator Settings	Value
Frequency	100 Hz
Amplitude	1.500 Vpp
Offset	1.500 VDC
Output Impedance	50 Ohm

PuTTY Output:

Parameter	RMS (V)	STD (V)	MIN (V)	MAX (V)	Faults
Test 1	2.630	0.652	1.465	3.300	BU:1 IN:1

Result: Burn-through and Instability faults detected. LED ON. MAX reached the ADC limit (3.300V) due to a 50 Ohm impedance mismatch.

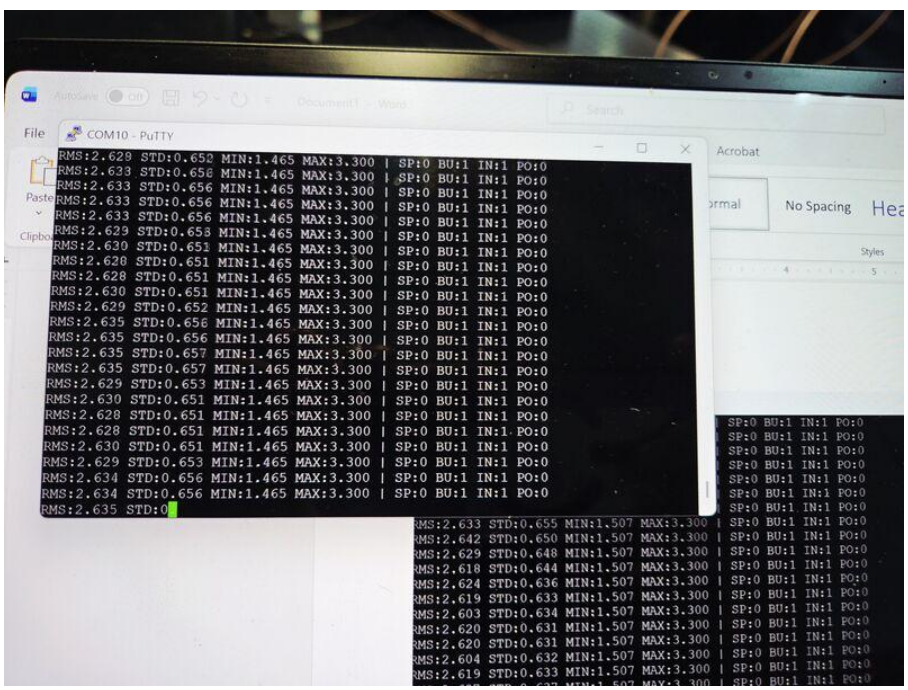

Fig 2: PuTTY showing BU:1 IN:1 faults

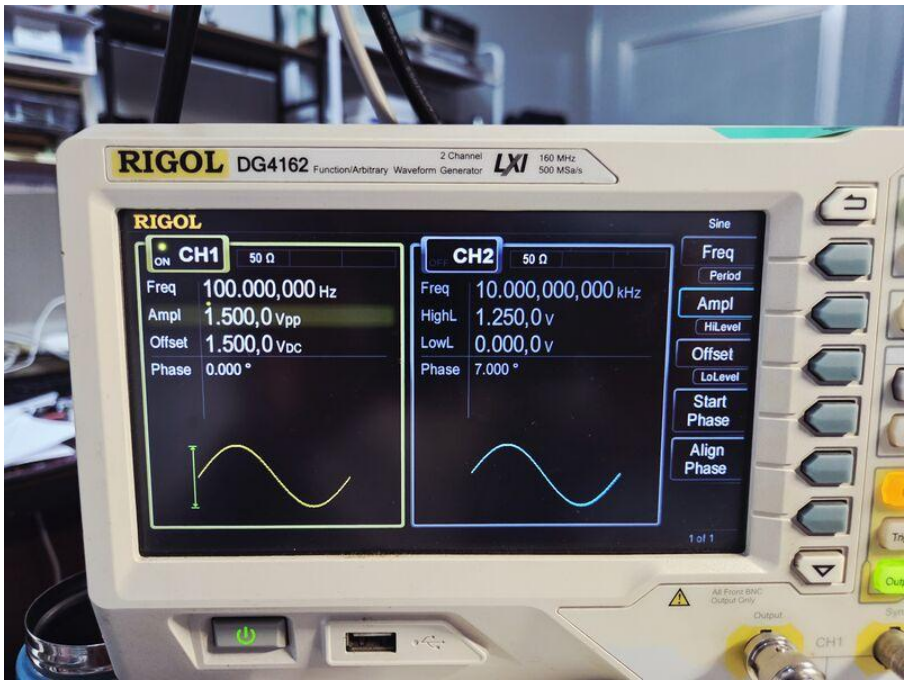

Fig 3: RIGOL DG4162 - 1.5Vpp, 1.5V offset, 50Ohm

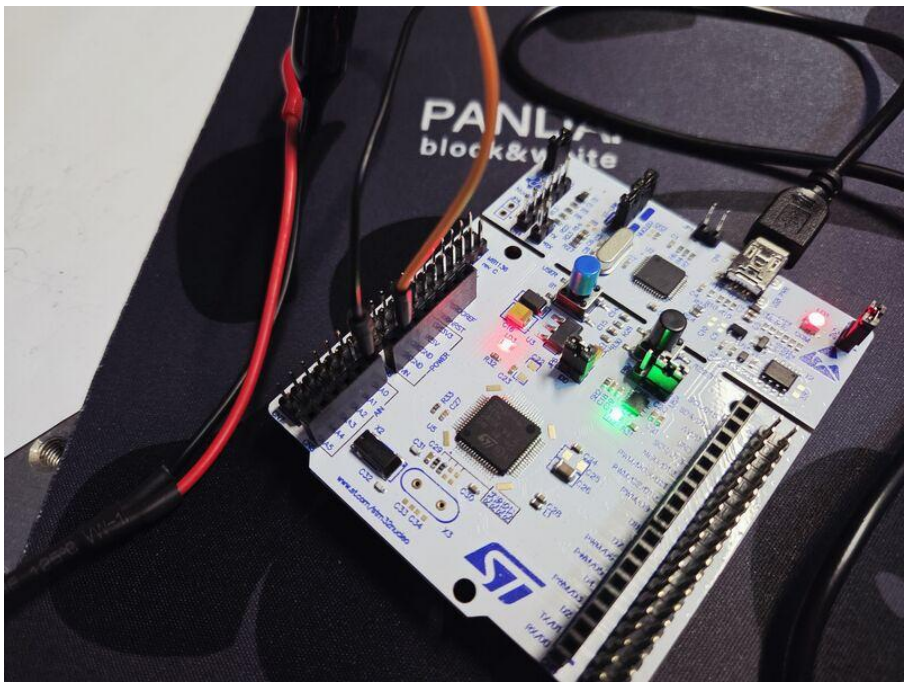
Fig 4: Nucleo board with LD2 LED ON (fault detected)

4.2 Test 2 — Normal Signal (No Faults) — After HighZ Fix

Function Generator Settings	Value
Frequency	100 Hz
Amplitude	200 mVpp
Offset	1.650 VDC
Output Impedance	HighZ

PuTTY Output:

Parameter	RMS (V)	STD (V)	MIN (V)	MAX (V)	Faults
Test 2	1.652	0.071	1.549	1.754	SP:0 BU:0 IN:0 PO:0

Result: No faults detected. LED OFF. The system correctly identified the normal weld condition after switching to HighZ impedance.

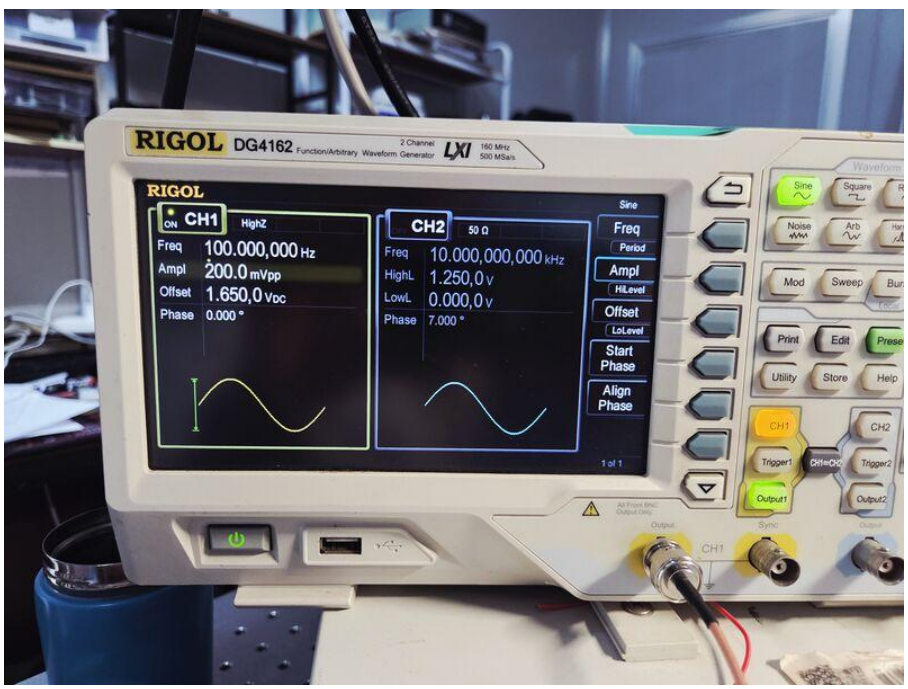

Fig 5: RIGOL DG4162 - 200mVpp, 1.65V offset, HighZ

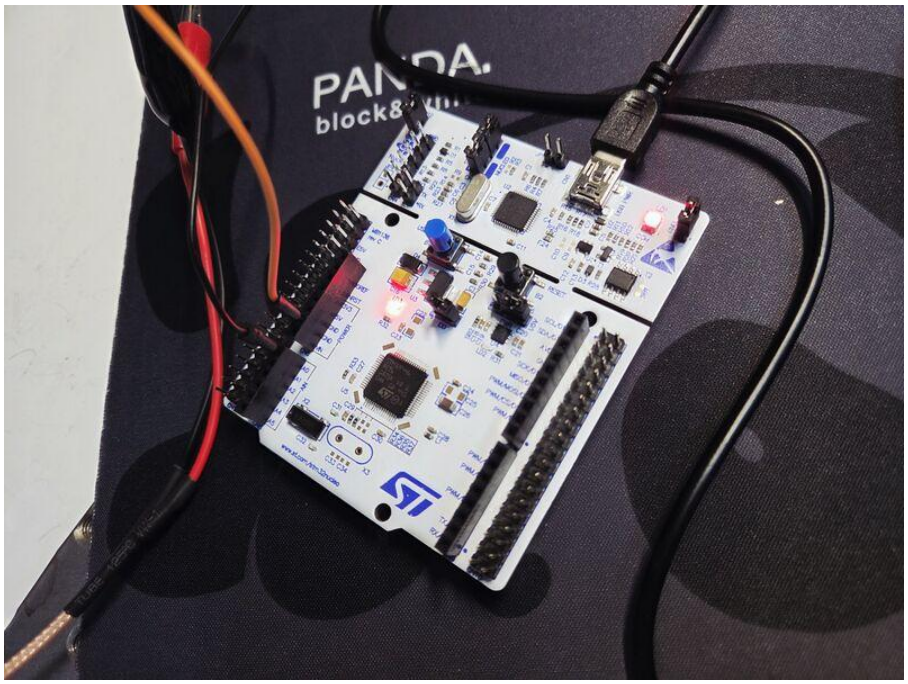

Fig 6: Nucleo board with LD2 LED OFF (no fault)

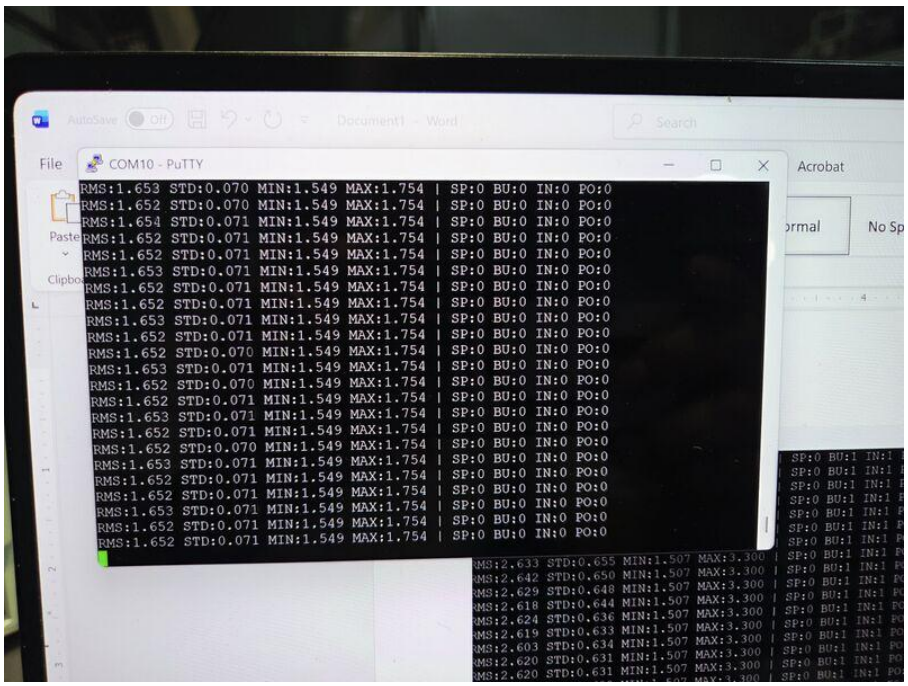

Fig 7: PuTTY showing SP:0 BU:0 IN:0 PO:0

4.3 Test 3 — Low Offset Signal (Porosity Detection)

Function Generator Settings	Value
Frequency	100 Hz
Amplitude	200 mVpp
Offset	300 mVDC
Output Impedance	HighZ

PuTTY Output:

Parameter	RMS (V)	STD (V)	MIN (V)	MAX (V)	Faults
Test 3	0.306	0.074	0.119	0.403	IN:1 PO:1

Result: Porosity fault detected (MIN = 0.119V < threshold 0.800V). Instability was also triggered due to the low signal range. LED ON.

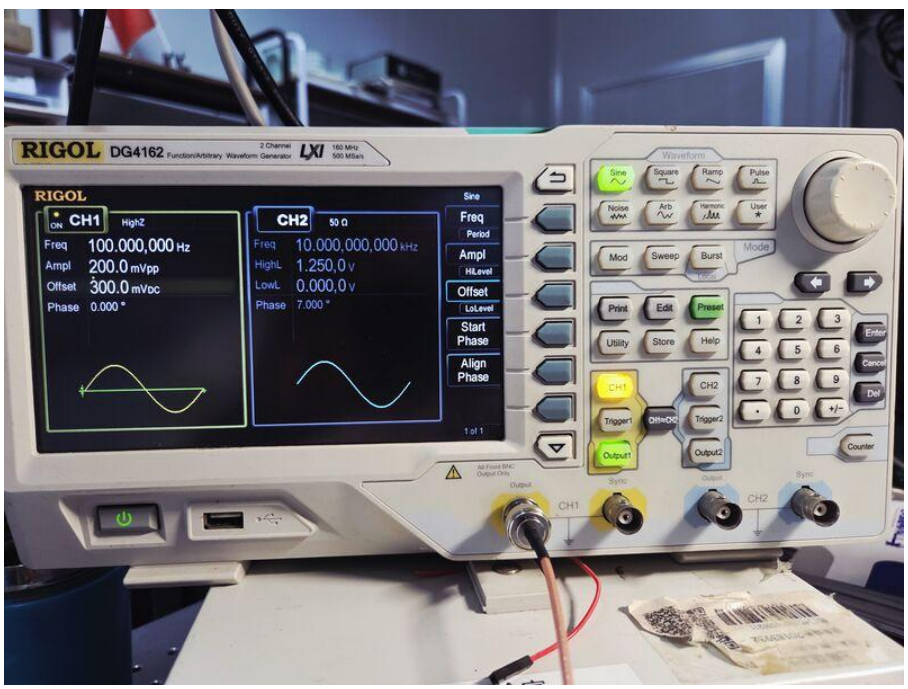

Fig 8: RIGOL DG4162 - 200mVpp, 300mV offset

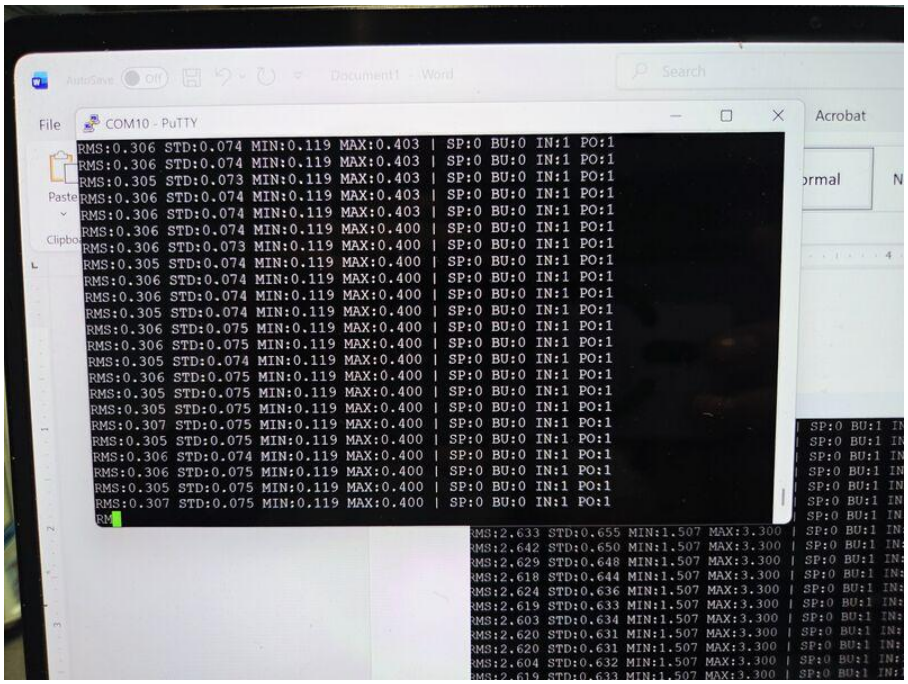

Fig 9: PuTTY showing IN:1 PO:1 faults

5. Test Summary

Test	Condition	Settings	LED	Result
Test 1	High Amplitude	1.5Vpp, 1.5V, 50Ohm	ON	BU:1 IN:1 detected
Test 2	Normal Signal	200mVpp, 1.65V, HiZ	OFF	All clear SP:0 BU:0 IN:0 PO:0
Test 3	Low Offset (Porosity)	200mVpp, 0.3V, HiZ	ON	IN:1 PO:1 detected

6. Key Learnings

- CubeMX + CubeIDE workflow generates correct HAL drivers and linker scripts automatically
- Software Packs (CMSIS-NN, CMSIS-DSP) must not be enabled unless required — they cause build failures
- -u _printf_float must be added to Linker flags (not Compiler) for float printf support
- Function generator output impedance (50Ohm vs HighZ) significantly affects the actual voltage at the ADC input
- ADC input must stay within 0V to 3.3V — negative or higher voltages damage the pin
- Circular buffer with 100 samples gives stable statistical measurements for fault detection

7. Conclusion

The Weld Monitor System was successfully implemented and tested on the STM32F446RE Nucleo board. The system accurately detected all four weld-fault conditions in real time using signal statistics computed from a sliding window of ADC samples. The UART output provided clear data logging, and the LD2 LED served as a visual fault indicator. All three test scenarios produced the expected results, validating the correctness of the detection algorithm and hardware setup.

End of Report